Partner API

- Evaluate the security requirements of an API of your choice and write a brief security requirements specification which mitigates against any risks associated with the API for enabling data sharing, scraping and connectivity between a program code written in Python and any of the following file formats/management systems (XML, JSON and SQL).

API stands for application programming interface. It is there to set out rules that allow software applications to communicate with each other enabling them to share data, features and even functionality.
API has a simplify application development that allows developers to integrate data, services and capabilities from other types of applications, this helps them as it stops them having to develop them from scratch.
API has multiple benefits including improved collaboration, accelerated innovation, data monetisation, end user security and privacy and system security.
Partner API connects business partners and usually developers accesses these in self service mode through something called an API developer portal. This still however needs to have login credentials to gain access.
API security works to prevent or mitigate attacks on APIs. To help prevent attacks API must follow protocols or rules set out, especially with partner API because two different companies are viewing and sharing data so there is a greater risk to it. These two companies will have to ensure they have credentials to access data that is only given to essential staff that need access. This falls under authentication so the credentials will know who is accessing the data, like an audit trial. Another protocol is encryption, if the data is encrypted in either one way encryption or mutual encryption which is a two way encryption. This ensures if the data is accessed by unauthorised personal there is another firewall stopping them from accessing the date. Another protocol is audits, logging and version so like I said before about credentials allowing an audit trail this falls under this protocol as well as ensuring the version is always up to date to allow us to see what data was at risk if there was a breach. Sharing as little as possible is also a protocol that can allow to be over cautious and protect the data. System protection with throttling and quotas is where access is restricted to a limited number of messages per second, this is to protect the backend system bandwidth. This will stop anyone trying the hack the system because they will be limited with the amount of time they have on the system. Data validation means to check everything before the server accepts, this helps prevent bugs. Infrastructure means keeping the network up to date. API fireballing, building a firewall will help to prevent hacking and give a chance to secure data before it is hacked into.

References:

Away blog. (2023) API security: 12 essential best practices. Available at: https://blog.axway.com/learning-center/digital-security/keys-oauth/api-security-best-practices/amp [Date Accessed 17th February 2024].

IBM. (2024) what is an application programming interface (API)?. Available at: https://www.ibm.com/topics/api [Date Accessed 18th February 2024].